COMP 472 Mini Project 1 - Short Reflection

Team Members

Name	Student ID
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Contribution of Each Team Member

Noemie Corneillier	Knowledge base loading and data validation; unit testing and debugging.
Kaoutar Machouat	Semantic search implementation using embeddings and cosine similarity; interactive conversation loop.
Yijia Xu	Sentiment analysis integration and escalation policy for strongly negative messages.

Difficulties Faced and How We Overcame Them

The main difficulty was understanding the AI concepts behind embeddings, semantic similarity, and sentiment analysis before implementing them. We overcame this by reviewing the concepts first, then testing each feature separately before combining them. Another challenge was dividing the work clearly between teammates, so we assigned each member one main component and reviewed the final program together. We also faced dependency and environment issues in VS Code because the project uses several libraries, so we created a requirements.txt file and tested installation in a clean environment. Finally, the planned GUI was harder to test during development, so we kept the core assistant logic separate from the command-line interface to make a future GUI easier to add.

Actual Development Time Per Component

Component	Time	Work completed
Knowledge Base Loading	0.5 h	Created knowledge_base.csv, added Concordia-related questions, loaded it with pandas, and validated columns/data.
Understanding Embeddings	1.5 h	Learned how text is converted into vectors so similar questions can be compared even with different wording.
Environment Setup	0.5 h	Set up VS Code and installed transformers, sentence-transformers, scikit-learn, numpy, and pandas.
Python Review	0.5 h	Reviewed file handling, classes, functions, error handling, and clean organization.
Semantic Search	2.5 h	Generated embeddings for stored questions and used cosine similarity to retrieve the closest answer.
Sentiment Analysis	2.5 h	Used the Hugging Face sentiment-analysis pipeline and tested positive, neutral, and negative messages.
Escalation Logic	0.5 h	Added the rule to recommend a human advisor when sentiment is negative with confidence above 0.9.
Conversation Loop	1.0 h	Built the interactive loop, quit command, and input validation for empty messages.
Testing and Debugging	2.5 h	Tested CSV loading, embeddings, retrieval quality, sentiment output, edge cases, and repeated questions.
Extra Improvements	1.5 h	Improved documentation, added more Q&A; pairs, organized files for GitHub, and kept the code GUI-ready.
Having Fun	Unlimited	Experimented with different questions and thought about future GUI improvements.

Overall, our actual development time was close to the estimated project timeline. We divided the work between learning the AI concepts, implementing the required components, testing the assistant, and improving the project organization for submission.